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Optimizing the configuration of electric vehicle charging piles in public parking lots based on a multi-agent model

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ABSTRACT

With the rapid development of electric vehicles, how to improve the charging efficiency of electric vehicles has become a challenge. The Chinese government has made great efforts to build charging piles. At present, the most popular construction mode is to build charging piles on a fixed proportion of spaces in existing parking lots. The proportions of charging piles recommended by the government, which is known as a one-size-fits-all strategy. Challenges have arisen with the growth of electric vehicles, such as how to keep hot charging piles from overheating. This paper mainly simulates the actual demand and optimizes the configuration of charging piles to reduce the uneven spatial distribution of charging demand, to improve the overall utilization efficiency of regional charging stations. In particular, based on agent-based simulation technology, a multi-agent system of the road network, vehicles, and charging stations is constructed to simulate the charging behavior in the mixed scenario of electric vehicles and traditional fuel vehicles. The optimization model aims to design the configuration of charging piles to minimize the sum of electric vehicle queueing time, gasoline vehicle queueing time, and vehicle transfer time to idle parking lots. The model is solved by the genetic algorithm. This paper takes the Wulin Square business district in Hangzhou as a real-world example. The simulation results show that by optimizing the number of charging piles, the objective function is reduced by 17.1% compared with the initial number of charging piles, which effectively improves the operation efficiency of the parking system. To this end, this paper predicts the electric vehicle charging demand in the city and optimizes the configuration of charging facilities, which is conducive to maximizing social benefits in the transportation system and has good practical engineering application value.

1. Introduction

Electric vehicles (EVs) have been proposed as a key technology to help cut down the massive greenhouse gas emissions from the transportation sector [1]. Compared with traditional fuel vehicles, EVs have two shortages: short mileage and long charging time. The rapid development of EVs also depends on the construction and configuration of charging facilities [2]. The Chinese government made great efforts to build charging piles [3]. At present, the main construction mode of charging piles is to build charging piles on a fixed proportion of parking spaces in existing gasoline vehicle (GV) parking lots. That is, the government decides on a fixed proportion of public parking spaces and rebuilds them as EV charging spaces regardless of the actual demand for charging. This method will not only

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lead to uneven charging demands such as long queues for charging piles in parking lots in popular areas; but also seriously hinder the popularization of EVs [4]. Therefore, how to simulate and predict the current uneven distribution of charging demand, evaluate the overall benefit of the system and optimize the configuration of charging piles has become an urgent problem to be solved.

Optimization of charging pile configuration in the parking lot refers to the process of effectively planning and adjusting the location, quantity, and type of charging piles in the parking lot to achieve the best charging service effect and resource utilization efficiency. Its goal is to meet the charging needs of parking lot users for EVs to the greatest extent through scientific and reasonable configuration, improve the utilization rate of charging piles, reduce user waiting time, and optimize the distribution of charging piles in parking lots to meet the needs of EV users changes and future trends.

Since the real world is an environment where EVs and GVs coexist, the actual charging demand has temporal and spatial characteristics, and a series of activities such as vehicle arrival, queuing, and charging need to be considered. In addition, traveler's behavior and parking behavior are complex and difficult to model [5,6]. How to predict and evaluate this process becomes a challenge. An agent-based model (ABM) is a simulation model that simulates real-world scenarios and provides a natural description of complex systems. The model can take into account the heterogeneity of individuals. Over the past decade, agent-based simulation (ABS) has been widely used in transportation to achieve accurate predictions of how drivers will react in various road conditions [7]. This study proposes an ABM that can accurately describe interactions between travelers, network traffic conditions, and charging facilities, simulate the real process of charging and parking, and conduct a systematic evaluation.

Many management tools are recognized to alleviate unbalanced charging problem. Among these, the optimal configuration of charging piles is an effective strategy [8]. Charging pile configurations may change drivers' parking choices, therefore, leading to better parking allocation and resource utilization. Based on the ABM, this paper proposes a simulation optimization method, which combines the charging demand prediction and the charging pile optimization configuration problem to maximize the system benefit. The method consists of two models: a charging demand simulation prediction model and a charging facility configuration optimization model. Two models are run in parts: the first step is to predict the parking and charging demands for different types of vehicles according to the traffic conditions in the parking system; In the next step, the optimization model will use predicted charging demand to adjust the number of charging piles optimized by the genetic algorithm.

The rest of this paper is organized as follows. After the introduction in this section, Section 2 provides a literature review of the related literature. Section 3 introduces the method of the EV simulation model based on the ABM in detail. The genetic algorithm is used to solve the charging facilities configuration optimization problem in Section 4. A numerical example is provided in Section5. Finally, the conclusions of this paper and future research are given in Section 5.2.1.

2. Literature review

The charging behavior of EVs is more complex than the refueling behavior of GVs. EVs charging behavior is highly random in time and space. Travelers will make different decisions depending on the environment, such as traffic conditions and charging prices. And environment is affected by travelers' choices. Therefore, it is difficult to build a mathematical model that can describe all types of individual behavior. Caliskan et al. [9] established a continuous-time homogeneous Markov parking model based on queuing theory to simulate the vehicle self-organizing network. Klappenecker et al. [10] provided a mathematical proof of the expected time for a vehicle to leave a parking lot. Bae et al. [11] proposed a spatiotemporal model based on the hydrodynamic traffic model and M/M/s queuing theory to predict EV charging demand at fast charging stations along highways. Kumarsinh et al. [12] proposed a model predictive control-based user maximum utility model. However, several parameters of these models, such as departure and arrival rates, are not suitable for practical applications. Therefore, alternative approaches have been proposed. Many recent studies have reported the progress of building ITS using agent technology, including parking and charging-guided information systems. Chou et al. [13] proposed a dynamic parking negotiation and guidance algorithm through an agent-based platform. Sun et al. [14] simulated the driver's reaction to parking VMS under various parking information by using agent-based simulation. Boudali et al. [15] proposed an intelligent parking system based on a multi-agent approach to provide real-time decision assistance to drivers considering their preferences. Tasseron et al. [16] evaluated the impact of different proportions of subscription users through proxy simulations. Mei et al. [8,17] construct an agent simulation framework to compare parking pricing and parking reservation and optimize parking space allocation. These studies show that agent-based simulation can effectively describe the dynamic movement, decision-making, and self-organization of traffic, and can effectively simulate the actual charging demand imbalance state.

Configuration optimization is an effective means of balancing charging demand, which has garnered extensive research attention. Cai et al. [18] analyzed hotspots of parking demand using trajectory data from taxis in Beijing as a crucial basis for selecting charging station locations. Xu et al. [19] analyzed spatial behavior characteristics of travelers based on the concept of travel chains and established a spatial charging demand model for electric vehicles. Building upon demand forecasting, an immune optimization algorithm based on weighted Voronoi was used to optimize the selection of charging station locations. Shao et al. [20] took into account both queueing time at charging stations and mileage constraints of electric vehicles to establish a charging station location model in a competitive environment, aiming to maximize the overall utility of users. An immune optimization algorithm was employed to solve the model. Wu et al. [21] proposed a Stochastic Flow Capture Location Model (SFCLM), which framed the optimization problem of charging station locations as a two-stage stochastic integer programming. In the first stage, station locations were determined, while in the second stage, the flow of electric vehicles between different travel chains was determined. Most configuration optimization models consider the impact on the city grid and ignore the characteristics of charging facilities, land use, and the impact on traffic on the surrounding road network. Further research is still needed on how to optimize the configuration of charging facilities from the perspective of the transport system.

The integration of simulation and optimization techniques is also commonly used in transport system optimization. Simulation-optimization iteration is a new field where simulation and optimization models can be iteratively updated with parameters and run iteratively until a termination condition is reached [22]. There are many examples of simulation optimization iterations available. García-Magariño et al. [23] used ABM to model the effects of different coordination strategies in route planning for electric vehicle drivers. He proposed a routing strategy that showed statistically significant improvements in reducing waiting times at charging stations and global travel times. The integration of simulation and optimization is a useful tool. Optimization models can provide valuable information about the system. The simulation model allows a systematic analysis of some practical factors that are not considered in the optimization model. The simulation does not end until a final solution is reached that is realistic for the overall operation.

In a word. This paper enriches and expands the research on the optimization of charging pile configuration based on existing research and describes the randomness and time-varying nature of the charging demand of EVs mathematically, aiming at the improvement of service performance. A simulation model for charging demand forecasting is constructed by analyzing the influencing factors. After that, the objective function based on time cost is constructed, and the intelligent algorithm is used to optimize the solution to adjust the configuration of different numbers of charging piles. Further feedback and improvement at the micro-level will make the configuration planning and operation management of EV charging facilities collaborative and optimized to maximize system benefits. This paper adopts the ABM, which is flexible, self-determined, and realistic, due to the lack of relevant data in the current situation and the difficulty in the realization of complex charging behaviors used by traditional modeling methods.

3. Agent-based EV charging simulation model

The essence of the agent computation is to make decisions on its behavior through a series of computations. The traffic problem is essentially the process of each individual (vehicle) involved in traffic moving and making decisions on the road network. Therefore, the vehicles driving in the road network can be regarded as agents with independent decision-making abilities. Each charging station can calculate the real-time number of idle piles, count the total charging load, and make relevant decisions, so it can also be regarded as an agent. The ABM can show the complex spatiotemporal relationship of traffic individuals (vehicles) in the road network and the decision-making process after the driver interacts with the environment [24].

3.1. The aim of the study

The purpose of the ABM is to simulate the operation results of EVs and GVs under different charging pile configurations in public

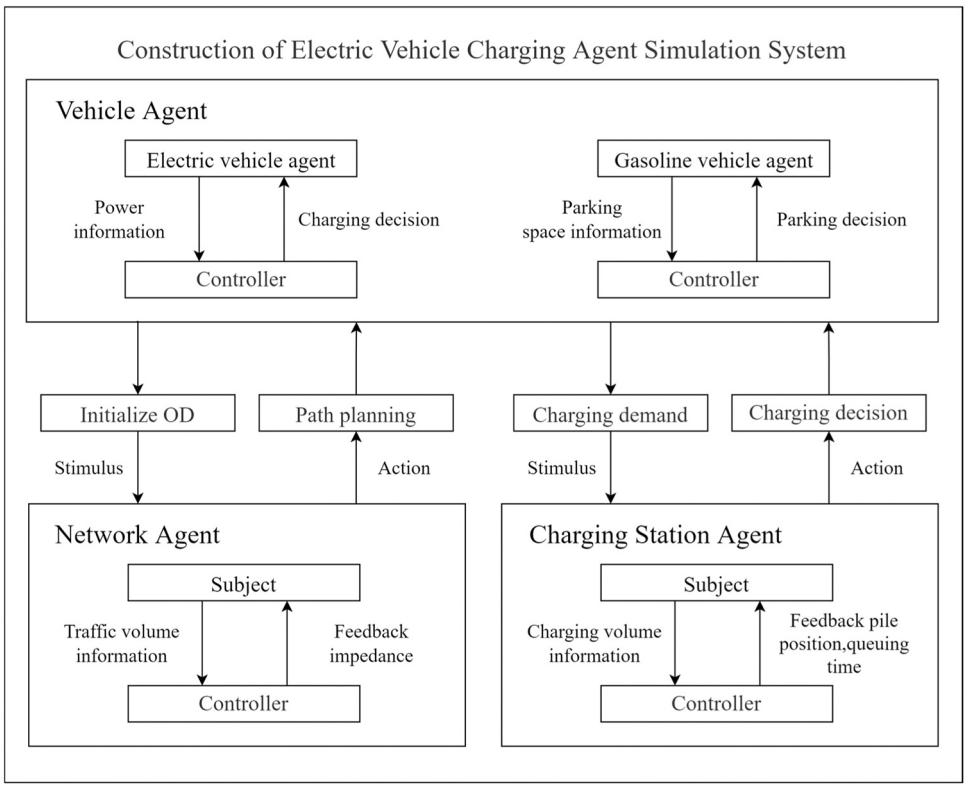


Fig. 1. Construction of electric vehicle charging agent simulation system.

parking lots. The primary outcome was measured as the total wait time for EVs at fast-charging stations and GVs in parking lots. In addition, through a configuration optimization model that takes the sum of the minimum queuing time of EVs and GVs and the time to transfer to alternative idle parking lots as the objective function, the optimization solution is carried out by the genetic algorithm. Finally, with the ABM, researchers can extend it to evaluate more detailed policies and richer representations of real-world scenarios.

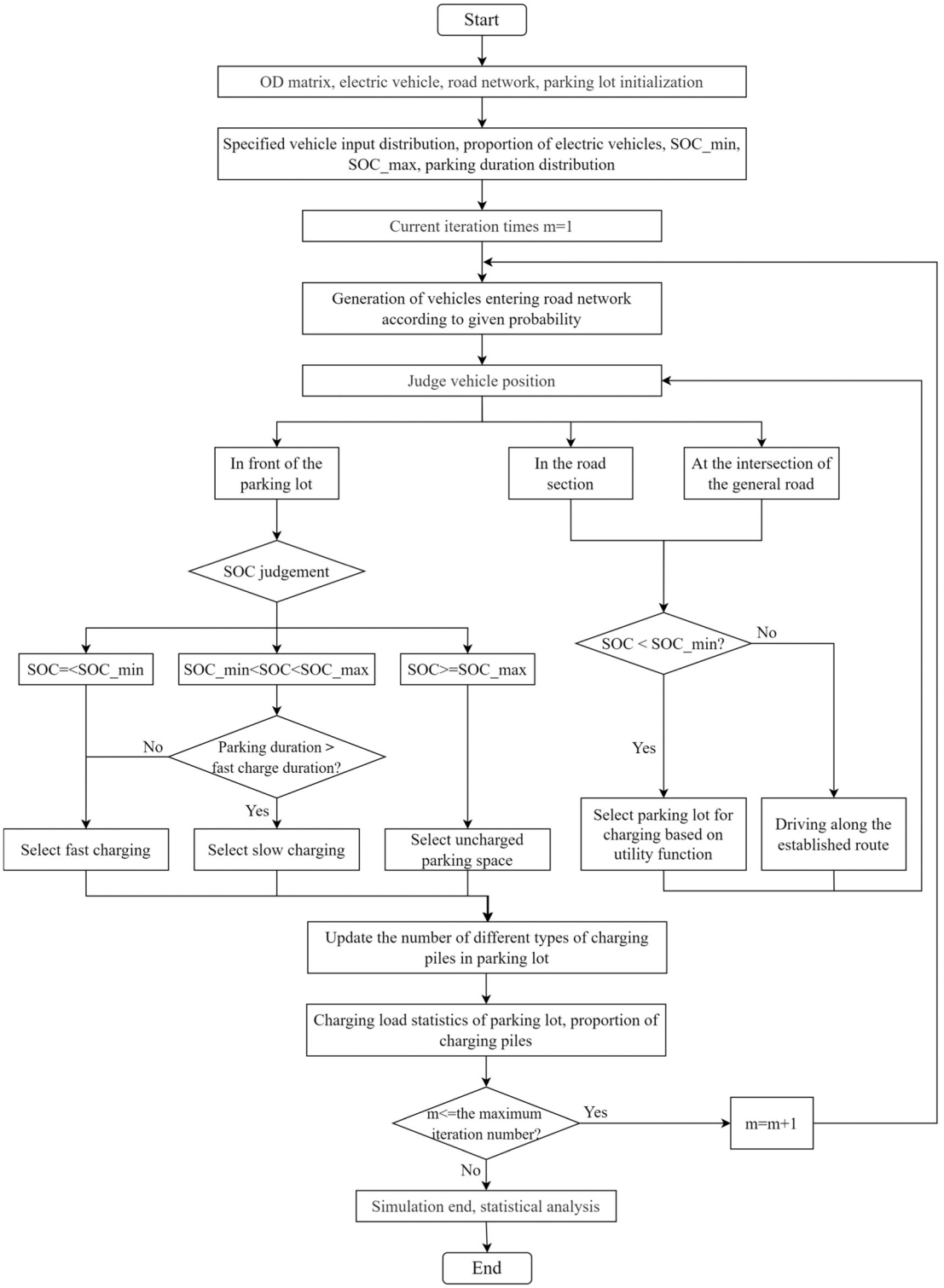


Fig. 2. Prediction of charging demand.

3.2. Agent attributes

The agent simulation system mainly includes three parts: the vehicle agent, the road network agent, and the charging station agent. The interaction and decision-making relationship between each agent is shown in Fig. 1.

The vehicle agents are composed of EV agents and GV agents. The EV agent's charging decision assumes that the primary consideration is the remaining power. When its power is less than the lower threshold of charging, the nearest charging station is selected for charging. The GV agent mainly considers its destination and parking behavior.

The road network agent mainly considers the traffic volume on each road section and plays an important role in the path decision of the vehicle agent by calculating its saturation, travel time, and other metrics.

The charging station agent mainly perceives the charging vehicle at a specific time and predicts the utilization rate of charging piles in the future. Then it calculates the queuing time of the vehicle to provide a decision-making basis for the charging of EVs.

3.3. Simulation process design

The agent simulation system constructed in this paper can realize functions such as charging demand prediction, post charging information, route planning, and configuration optimization of EV charging piles. This section will mainly introduce the design of the first three functions, configuration optimization of EV charging piles will be introduced in the next chapter.

3.3.1. Charging demand prediction

The idea of charging demand prediction is to calculate the number of slow charging piles and fast charging piles that could meet for all EVs in each parking lot according to the travel origin and destination, travel power, and travel purpose of EVs (reflected in the parking time). The detailed prediction steps are shown in Fig. 2.

The specific steps are as follows.

Step 1: Initialize parameters.

3.4. Initialize the simulation road network

The actual map in the road network is selected to obtain the road network agent topology structure. Set the starting node-set $\boldsymbol{O}$, the destination node-set $\boldsymbol{D}$, the parking lot node-set $\boldsymbol{P}$, the intersection node $\boldsymbol{C}$, and the road segment set $\boldsymbol{S}$. Initialize the connection attributes and distances between various nodes. Initialize the connection attributes and distances between various nodes. Initialize the road segment level (arterial road/secondary road), traffic capacity C_s, free flow speed v_s^0. Preload the initial flow of each road segment in the road network $q_s(0)$, then use the Bureau of Public Road (BPR) function to calculate the travel speed $v_s(0)$ and travel time $t_s(0)$ of each road segment of the initial road network, which are calculated as follows:

$$v_s(0) = v_s^0 \cdot \left(1 + \alpha_s \left(\frac{q_s(0)}{c_s}\right)^{\beta_s}\right) \tag{}$$

$$t_s(0) = \frac{L_s}{v_s(0)} \tag{2}$$

where α_s, β_s are the parameters of the BRP function in the road segment s, L_s is the length of road segment s.

3.5. Initialize charging stations and parking lots

Initialize the position P_i of each parking lot, the capacity C_p, the number of fast charging piles $N_{fast}(P_i)$, the number of slow charging piles $N_{slow}(P_i)$, the number of ordinary parking spaces $N_{com}(P_i)$, $L_{p\to c}$ is the walking distance from the parking lot p to the destination c.

3.6. Initialize and generate vehicles

According to the travel OD, the vehicle agent is randomly generated to enter the road network. ID_c is the vehicle number, O_c is the vehicle's orientation, D_c is the vehicle destination, t_c^{birth} is the vehicle generation time. The initial path planning is carried out according to the Dijkstra algorithm, the intersection B_c that the vehicle will pass through is determined by the adjacency relationship between the starting point and the intersection of the road network.

Among the generated vehicles, a certain percentage of vehicles are randomly selected as EVs, and the EV label f_c^r is 1. The GV label f_c^r is 0 and the EV is given an initial state of charging $SOC(i)$.

3.7. Initialize the charging properties

Initialize the proportion of EVs and the distribution of the parking time of all vehicles. Initialize the lower threshold of charging $SOC_\min$ of EVs. The meaning of this value is that if the power of the EV is lower than this value, it needs to find a charging station to

charge immediately. Initialize the upper threshold of charging $SOC_\max$, the meaning of this value is that if the EV power is higher than this value, it does not need to charge.

3.8. Initialize the simulation environment

The simulation time horizon is 24 h and the simulation interval $\Delta t = 1\min$. The current iteration number $m = 1$. The maximum iteration number $M = 1440$.

Step 2: Determine the position of the vehicle and the remaining power.

For each vehicle on the road network, its location on the road network has three states: on a road segment, at a road intersection, and in front of a parking lot. The position of the vehicle and the remaining power is calculated in each step of the simulation.

Step 3: Update the charging station information.

3.9. Charging decisions of EV agents at parking lots

When an EV is located in front of the parking lot, its power will be judged. If the power is lower than the lower charging threshold, the vehicle will choose fast charging. At the same time, the demand for fast charging piles in the parking lot will increase by 1.

If its power is more than the lower charging threshold but less than the upper charging threshold, compare the parking time with the estimated fast charge time of the parked vehicle. Under the assumption of fast charging rules (the vehicle must leave when it's fully charged), if the parking time is longer than the expected fast charging time, the EV chooses slow charging to avoid moving the car, and the demand for slow charging piles in the parking lot increases by 1; On the opposite, the EV chooses fast charging and the demand for fast charging piles in the parking lot increases by 1.

If the remaining power is more than the upper charging threshold, the EV will choose not to charge and the number of GV parking spaces will increase by 1.

3.10. Charging decisions of EVs on-road sections or intersections

When an EV is on a road section or at an intersection, the power iscalculated. If the power is more than the lower charging threshold $SOC_\min$ the vehicle will travel to the initial destination according to the planned route. If the power is less than the lower charging threshold, the user selects the parking lot for charging based on the negative utility function. The parking selection model considers factors including travel time, walking distance, parking fees, etc. The calculation of the negative utility function is as follows:

$$U^r_{bpc}(m) = a_1 \cdot t_{bp}(m) + a_2 \cdot d_{pd} + a_3 \cdot f_p + a_4 \cdot \zeta \tag{3}$$

Among them, $U^r_{bpc}(m)$ represents the parking selection utility of the reserved vehicle c from the current intersection b to the parking lot p in the m-th iteration. $t_{bp}(m)$ is the shortest travel time from the current intersection b to the parking lot p in the m-th iteration (obtained by the Dijkstra algorithm). d_{pd} is the walking distance from the parking lot p to the destination d. f_p is the parking fee. ζ is a random item. a_1, a_2, a_3, and a_4 are the important coefficients of each factor and they are all negative [25,26].

3.11. Update the demand for charging piles when the vehicle leaves

For fast-charging vehicles, when the battery is fully charged they will drive away and the required number of fast-charging piles in the parking lot will be updated accordingly.

For slow-charging vehicles, they will leave at the end of the parking time according to the initial parking time and the demand for slow-charging piles in the parking lot will be updated at the same time.

Step 4: Analyze and count the charging demand of the parking lot.

When $m = 1440$, the simulation ends. Calculate the demand for fast charging and slow charging.

3.11.1. Post charging information

In the initial charging demand prediction, we assume that EVs go to the parking lot charging stations to charge without the limitation of the number of charging piles. When optimizing the configuration of electric vehicle charging piles, it's necessary to consider the limited number of charging piles in the parking lot. We assume that the charging information can be shared with EVs in real-time to provide decisions for charging decisions and path planning.

3.11.2. Route planning

The charging demand prediction is used to set the capacity of the charging piles of the charging station. According to the predicted demand and the actual situation of the parking lot, the actual number of charging piles in the parking lot can be calculated. Therefore, the path planning can be re-planned for EVs according to the remaining number of charging piles or the estimated queuing time at the limited charging piles. The precondition for path planning is based on the sharing of charging information at all charging stations.

The detailed steps of route planning are as follows.

Step 1: Calculate whether the remaining power can support the next trip.

According to the travel OD, calculate whether the remaining power meets the next travel demand. If not, then find the nearest

charging station and follow the shortest route to charging. Otherwise, search for the ID_p of the parking lot near point D and go to step 2 of route planning. To meet the next travel demand means that the remaining power is still more than 0 after arriving at the destination charging station so that the EV can be charged at the destination charging station.

Step 2: Charging decision at destination.

Calculate the remaining power of the EV after reaching the destination and make a judgment according to the distribution of the parking time. The judgment principle is the same as the charging deciding process in Section 3.3.

Step 3: Calculate the charging pile utilization rate of the destination charging station.

Calculate the charging pile utilization rate of the charging station ID_p. If there are remaining charging piles, the path planning is carried out according to the Dijkstra algorithm after considering the road resistance through the BPR function. If the charging piles are not available, go to step 4.

Step 4: Charge induction.

Under the condition that there is no charging pile at the destination charging station, the EV searches for the nearest charging station with the remaining piles. Take the time as the optimization target. If the sum of the driving time to a charging station with remaining piles and the walking time from the charging station to the destination is less than the estimated queuing time, then the EV navigates to the parking lot. Otherwise, it navigates to the nearest parking lot of the destination.

4. Configuration optimization of EV charging piles

Based on charging demand prediction, the proposed ABM is used to optimize the configuration of charging facilities.

4.1. Determine the initial construction proportion of charging piles

According to the charging demand prediction in the previous chapter, the maximum demand of fast charging piles and slow charging piles of each parking lot charging station is obtained. Combine the prediction result and parking lot capacity to determine the initial proportion of the lot to be rebuilt as EV charging piles.

4.2. Electric vehicle charging decision

Different from the charging demand, the charging demand forecast only considers the EV's power and parking time to decide fast charging, slow charging, or not charging. At the stage of optimization configuration, the main limiting factor is the capacity of the charging station. In simple terms, for vehicles with charging needs, the cost structure mainly includes charging waiting time and the time to find parking lots for remaining charging piles; for vehicles with parking needs, the cost structure mainly includes parking waiting time and time to find parking lots for remaining parking spaces. By comparing the time of waiting and searching, choose the solution with less time cost.

First of all, determine the remaining power of the EV.

(1) Condition with $\boldsymbol{SOC < SOC_min}$.

If the initial remaining power SOC is less than the lower charging threshold $SOC_\min$, it is assumed that the EV selects fast charging. The EV goes to the target parking lot and makes a judgment on the remaining number of charging piles in the parking lot before arriving at the parking lot.

If there is a fast-charging pile, the vehicle will choose fast charging; otherwise, the car has two options. Option one is to queue up in the target parking lot and calculate the number of vehicles queued for fast charging and estimated queuing time.

a) Calculate the time for the EV to be fully charged:

$$X_FT_i = \frac{(100 - SOC_i) \cdot FT_i}{100} \tag{4}$$

X_FT_i ——The estimated time for EV to be charged quickly to full (min).

FT_i ——The theoretical time for the car to be charged from zero to full (min).

SOC_i ——The real-time charge of the i-th vehicle that is charging when the n-th queued vehicle arrives.

b) Sort the charging time of the EV ascendingly. Suppose the car in the N-th position is the C-th car. The estimated queuing time of the N-th queuing EV is:

$$w1_Fast = X_FT_c \tag{5}$$

Option two, choose a nearby parking lot with remaining charging piles for charging. The cost of this decision mainly consists of two parts. The first part is the transfer time from the target parking lot to the new parking lot and the second part is the walking time from the new parking lot to the destination.

c) Sum of time cost:

$$w2_Fast = T^N_{transfer} + \alpha \cdot T^{Nj}_{walk} = \frac{L_{ij}}{V_0} \cdot \frac{60}{1000} + \alpha \cdot \frac{L_{Dj}}{60 \cdot V_{walk}} \tag{6}$$

$w2_Fast$ ——The total time cost of EV N to transfer from target charging station i to charging station j and then walk back to destination d.

$T^{N}_{transfer}$ ——The time for the electric vehicle to be transferred from charging station i to charging station j (min).

L_{ij} —— The least impedance path to transfer from charging station i to charging station j.

V_0 ——The average speed of vehicles in the city.

T^{Nj}_{walk} ——The time for the EV driver to walk from the j charging station to the destination (min).

L_{Dj} ——The shortest path from charging station j to the destination (m).

V_{walk} ——The average walking speed of the driver (m/s).

α ——Walking resistance coefficient. Considering EV drivers have low willingness to walk, multiply a walking resistance coefficient before the walking time to increase the cost of walking travel time.

Calculate the time cost of the two options separately and choose the option with the lowest cost.

(2) Condition with $\boldsymbol{SOC > SOC_max}$.

If the initial remaining power SOC is larger than the upper charging threshold $SOC_\max$, then the EV does not need to be charged at this time. The main purpose is to make a parking decision. The EV goes to the target parking lot and makes a judgment on the remaining number of charging piles in the parking lot before arriving at the parking lot.

If the parking space is available, the vehicle will be parked in the target parking lot; if there is no parking space, the vehicle will have two options. Option one is to queue up in the target parking lot and calculate the number of queued parking vehicles and the estimated queuing time. Option two is to choose a nearby parking lot with remaining parking spaces for parking. Calculate the cost of the two options separately, and choose the one with the lower cost.

(3) Condition with $\boldsymbol{SOC_min < SOC < SOC_max}$.

If the initial remaining power is between the lower charging threshold $SOC_\min$ and the upper charging threshold $SOC_\max$, the EV will use the parking time distribution and the remaining number of charging piles at the charging station to make a comprehensive judgment.

First, judge whether the charging station has a slow charging pile, and if so, choose slow charging; otherwise, judge whether there is a GV parking space. If so, choose a GV space. If not, judge whether there is a fast-charging pile and decide according to the parking time. If the parking time is less than the fast-charging full time then select the fast charging. Otherwise, there are three options. Option one is to estimate the queuing time at the slow charging piles.

a) Calculate the time for the EV to be fully charged:

$$queue_ST_c = slow_PT_c + arrive_ST_c - m \tag{7}$$

$queue_ST_c$ ——The time at which the slow-charging EV c leaves the charging station.

$slow_PT_c$ ——The initial parking time of EV c.

$arrive_ST_c$ ——The charging start time of EV c.

m ——The current time step.

b) Sort by departure time ascendingly:

$$Slow_T = sort(queue_ST_c) \tag{8}$$

c) The queuing time of EV i is the departure time of the charging vehicle in the nth position:

$$w1_Slow = Slow_T_n \tag{9}$$

Option two is to estimate the queuing time of GV parking spaces.

a) Calculate the parking time of all parked cars, which is similar to that of slow charging:

$$queue_NT_c = norm_PT_c + arrive_NT_c - m \tag{10}$$

$queue_NT_c$ ——The time to leave the stopped GV c after the parking.

$norm_PT_c$ ——The initial parking time of the GV c.

$arrive_NT_c$ ——The starting moment of the parking of the GV c.

m ——the current moment.

b) Sort by departure time ascendingly:

$$Norm_T = sort(queue_NT_c) \tag{11}$$

c) The queuing time of EV i is the departure time of the charging vehicle in the nth position:

$$w1_Norm = Norm_T_n \tag{12}$$

Option three is to select a nearby parking lot with remaining parking spaces for charging or parking. The calculation method is the same as above. Finally, choose the lowest cost option.

4.3. Parking decisions of GVs

The parking decision of a GV is mainly based on the capacity of the parking lot. Before the GV arrives at the parking lot, it first determines whether there is a parking space. If so, it will park in the target parking lot. If not, then there are two options. First, queue

up in the target parking lot and calculate the number of parking vehicles in the queue and the estimated queue time. The calculation method is the same as the formula (10,11,12). Second, select a nearby parking lot with remaining parking spaces for parking. The calculation method is the same as the time cost of finding a new charging station. Calculate the cost of the two options separately and choose the option with the lowest cost.

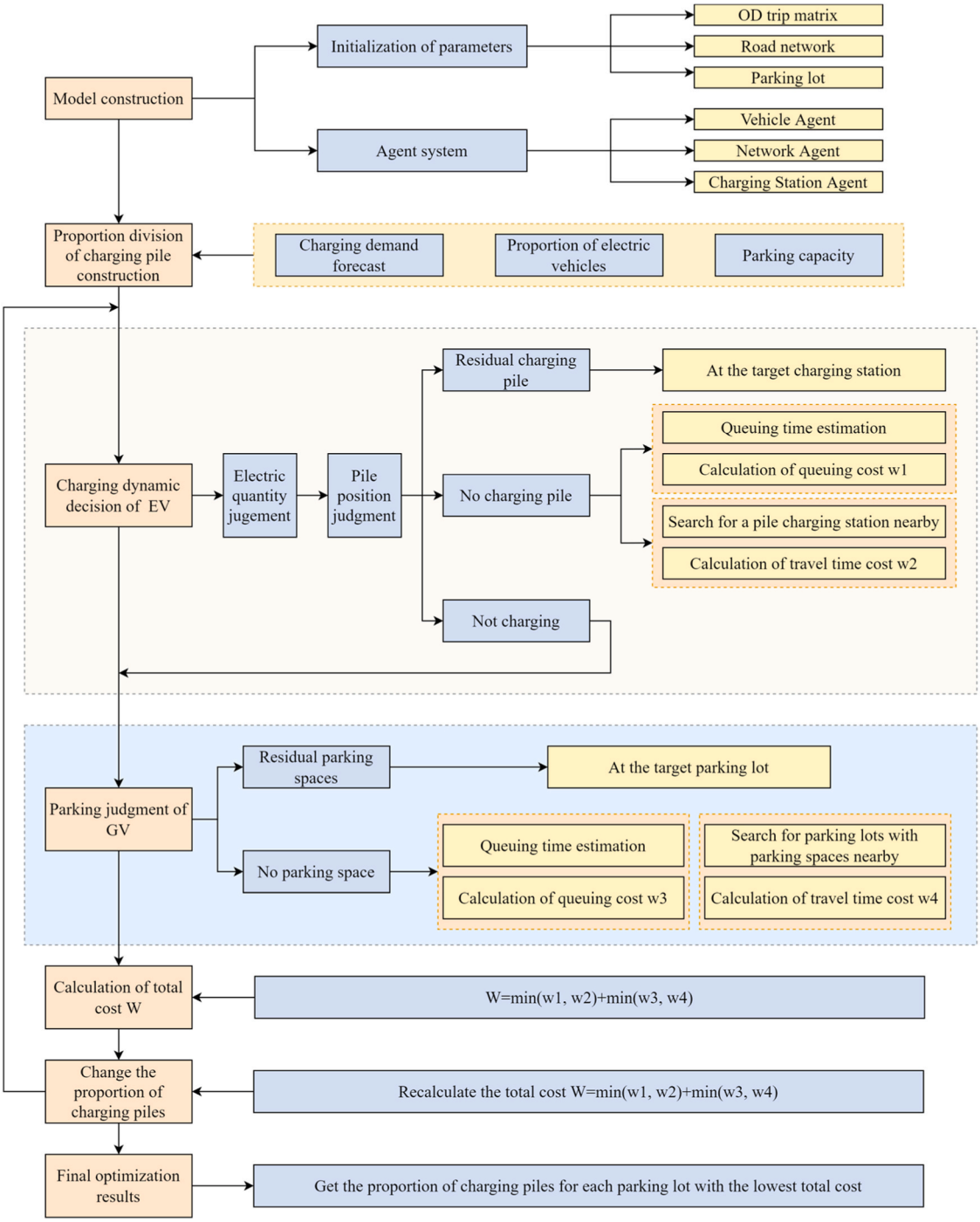


Fig. 3. The technical route for optimizing EV charging facility configuration.

4.4. Update the queue

After the vehicle makes a choice based on the minimum time cost, update the occupied number of common parking spaces, fast charging piles, and slow charging piles in the parking lot, as well as the number of queued vehicles. Vehicles that have completed parking and charging are removed from the parking lot, and the vehicle ends its simulation.

4.5. Construct the objective function

The time cost of various types of vehicles mainly includes the time of queuing and finding new charging stations or parking spaces. For fast charging:

$$W_Fast = \min\left(\begin{array}{c}\dfrac{(100 - SOC_i)\cdot FT_i}{100},\\ \left(\dfrac{L_{ij}}{V_0}\cdot\dfrac{60}{1000} + \alpha\cdot\dfrac{L_{Dj}}{60\cdot V_{walk}}\right)\end{array}\right) \tag{13}$$

For slow charging:

$$W_Slow = \min\left(\begin{array}{c}(slow_PT_c + arrive_ST_c - m),\\ \left(\dfrac{L_{ij}}{V_0}\cdot\dfrac{60}{1000} + \alpha\cdot\dfrac{L_{Dj}}{60\cdot V_{walk}}\right)\end{array}\right) \tag{14}$$

For GVs:

$$W_Norm = \min\left(\begin{array}{c}(norm_PT_c + arrive_ST_c - m),\\ \left(\dfrac{L_{ij}}{V_0}\cdot\dfrac{60}{1000} + \alpha\cdot\dfrac{L_{Dj}}{60\cdot V_{walk}}\right)\end{array}\right) \tag{15}$$

Finally, the sum of the time costs of all vehicles is obtained:

$$W = \sum_{i=1}^{n} W_Fast + \sum_{i=1}^{m} W_Slow + \sum_{i=1}^{Q} W_Norm \tag{16}$$

In the formulas: n is the number of fast-charging EVs in the area accumulated throughout the day. m is the number of slow-charging EVs in the area accumulated throughout the day. Q is the total number of parking vehicles in the area accumulated throughout the day.

The optimization goal is to minimize the value of the objective function W by adjusting the number of fast charging piles, slow charging piles, and the number of GV parking spaces in each parking lot.

4.6. Change the number of charging piles in each charging station

Determine the upper limit of the proportion of charging piles in each charging station and change the number of charging piles in each charging station. It is more computationally intensive to find the optimization solution directly by enumeration comparison, so this section adopts the commonly used heuristic algorithm—genetic algorithm [27] to solve the optimization objective function. In the real operation, because the behavior of electric vehicles has certain randomness, so the solution result based on a genetic algorithm can make a relatively better plan for the charging station management to set the number of charging piles in the parking lot. For the arrangement of the number of fast charging posts, the arrangement of the number of slow charging posts, and the arrangement of general parking spaces in each parking lot, three simulations are conducted, and the average value of the optimization objective is taken as the optimization result to reduce the volatility of a single simulation and enhance the robustness of the program. After simulating all cases, the charging pile configuration scheme with the smallest objective function among them is obtained. To obtain the optimization effect, it can be evaluated by comparing the sum of travel time cost of all vehicles before and after optimization.

The technical route for the optimized configuration of charging facilities for charging vehicles is shown in Fig. 3 below.

5. Case analysis

To verify the effect of the EV charging pile configuration optimization scheme proposed in this paper, we take the typical Central Business District (CBD) Wulin business district in Hangzhou as a real-world case study. The ABM is constructed to calculate the charging demand of each parking lot in the case of different proportions of EVs and optimize the configuration of charging piles in nine parking lots in the study area.

5.1. Initialization of the road network agent

Wulin business district is the core area of Hangzhou's central urban area. It is a modern commercial shopping center in Hangzhou with commercial and residential functions. It's also a cultural center and a public transportation hub distribution center. The abstract

simulated road network topology is shown in Fig. 4.

The simulated road network includes 4 departures, 4 destinations, 18 intersections, 9 off-street parking lots, and 54 road sections.

This paper sets the total OD amount as 2850, and the parking space capacities of each parking lot are 200, 180, 150, 160, 180, 150, 150, 200, and 200, the initially occupied berth are 40, 45, 30, 55, 55, 55, 55, 55, and 55. The initial remaining power of the EVs follows a normal distribution with a mean value of 50% and a variance of 25%, and the parking time of the vehicle follows a normal distribution with a mean value of 120 min and a variance of 80 min. The lower charging threshold $SOC_\min$ is 30% and the upper $SOC_\max$ is 70%. The estimated time X_FT from 0 to full charging of the EV is set to $0.7h$ and the estimated time X_ST from 0 to full charging of the electric vehicle is set to $9.5h$. The walking impedance coefficient α is set to 2 for fast charging and 1.5 for slow charging. The pedestrian walking speed is set to 1 m/s. The average speed of the vehicle is 9 m/s.

In this section, the initial value of the genetic algorithm population is set to the initial fast charging pile configuration number Cap_FP and the initial slow charging pile configuration number Cap_SP obtained by the charging demand prediction. The crossover rate is 0.85 and the mutation rate is 0.15. The population number is 20 and the iteration generation is 50. Taking W as a fitness function for simulation optimization.

This paper uses Matlab to package the ABM into a function. The input parameters are the combination of the number of fast charging piles, the number of slow charging piles, and the number of GV parking spaces in each parking lot. The output parameters are the optimization target value. To reduce the volatility of the simulation results, each group of parameters is simulated three times and averaged. Finally, it's optimized and solved by using the Matlab genetic algorithm toolbox.

5.2. Optimization results

The charging demand of 9 parking lot is predicted and the accumulated number of charging vehicles in each charging station is analyzed under the condition of different EV ownership according to the previously designed parameter. It is calculated that the simulation area needs a total of 52 fast charging piles, 101 slow charging piles and the total capacity of the parking lot is 1570. Therefore, the initial trial construction proportion of fast charging piles in the area is 3%, the proportion of slow charging piles is 6% and the total proportion of charging piles is 9%, which are as shown in Table 1 below.

The initial demand for fast charging piles, slow charging piles, and parking spaces at each charging station are calculated and shown in Table 2.

5.2.1. Calculation results under the initial construction proportion

If the vehicle is parked or charged in the target parking lot according to the initial OD, then the time cost is counted as 0. If the vehicle fails to park or charge, then the cost of queuing and transferring to the new parking lot will be calculated. These two kinds of costs are used as the basis for decision-making for queuing or transferring to the target parking lot. The time cost and other related results are calculated as follows with the number of initial fast-filling piles $Cap_{FP} = [5, 6]$ and slow-filling piles $Cap_{SP} = [9\text{–}12]$.

First, we calculate the queuing time cost $w1$ of each vehicle and the time cost $w2$ of transferring to the new parking lot under the initial number of charging piles of each parking lot obtained from the charging demand prediction in Chapter 3. For the convenience of analysis, the original time cost data was sorted (the same as below). The calculation is shown as the red line in Fig. 5 and Fig. 6.

As can be seen from the red line in Figs. 5 and 6, the queuing time cost varies from 0 to 50 min and the time cost of transferring to the new parking lot varies from 0 to 18 min. $w2$ increases in a step-like manner compared with the exponential growth of $w1$. This is caused by the transfer of many vehicles to the same new parking lot because the target parking lot has no spaces to park or piles to charge.

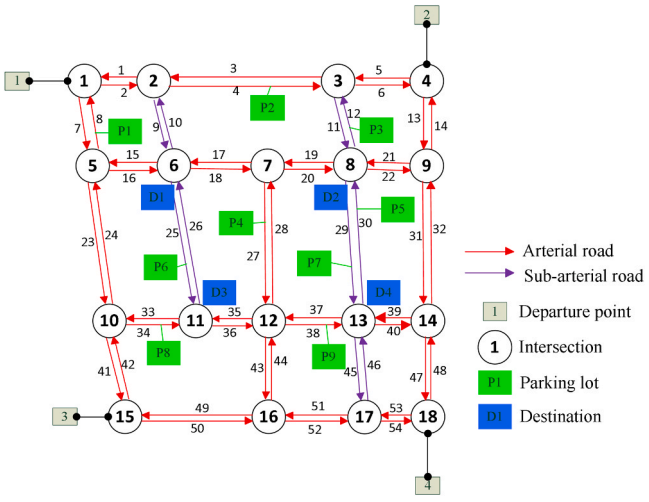


Fig. 4. The abstract simplified simulation road network.

Table 1
The proportion of charging pile demand and construction.

Parking lot ID	Demand for fast charging piles	Demand for slow charging piles	Parking capacity	Proportion of fast charging piles	Proportion of slow charging piles	Proportion of charging piles
1	10	10	200	5.0%	5.0%	10.0%
2	4	19	180	2.2%	10.6%	12.8%
3	3	0	150	2.0%	0.0%	2.0%
4	4	20	160	2.5%	12.5%	15.0%
5	5	1	180	2.8%	0.6%	3.3%
6	3	12	150	2.0%	8.0%	10.0%
7	3	0	150	2.0%	0.0%	2.0%
8	14	16	200	7.0%	8.0%	15.0%
9	6	23	200	3.0%	11.5%	14.5%
Total	52	101	1570	Average proportion		9.4%

Table 2
The initial number of charging piles in each charging station.

Parking lot ID	Parking capacity	Number of fast charging	Number of slow charging	Number of GV spaces
1	200	6	12	182
2	180	5	11	164
3	150	5	9	137
4	160	5	10	146
5	180	5	11	164
6	150	5	9	137
7	150	5	9	137
8	200	6	12	182
9	200	6	12	182
Total	1570	47	94	1429

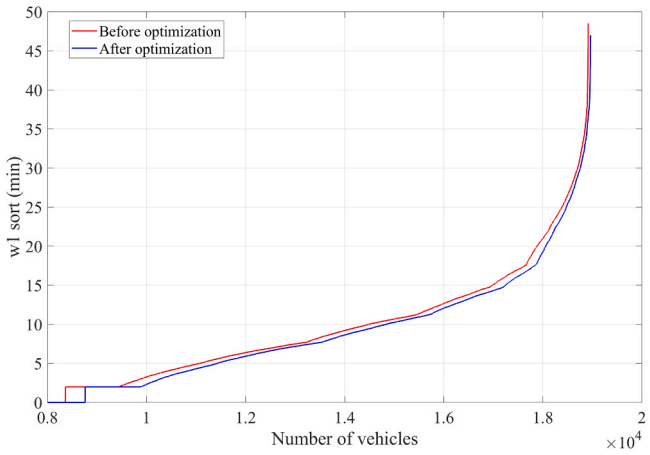


Fig. 5. The queuing time cost $w1$ of each vehicle before and after optimization.

We calculate $w1$ and $w2$ according to the vehicle in the parking lot. The total time cost of each vehicle is obtained as shown in Fig. 7.

It can be seen from Fig. 7 that during the simulation step size of 1440 steps, a total of 18745 vehicles are generated, of which 8171 vehicles are parked or charged according to the initial OD. It accounted for 43.6% of the total number of vehicles. The number of queued or transferred vehicles is 10,574, which are accounted for 56.4% of the total number of vehicles.

We sum the total time cost of each vehicle. The sum of the objective function W is obtained as:

$$W = 8.58 \times 10^4 (\text{min})$$

To further study the choice of vehicles that park in the target parking lot or fail to charge, this paper calculates the difference between $w1$ and $w2$. If $w1$ is greater than $w2$, the vehicle selects to transfer. Otherwise, the vehicle selects to queue. We statistically obtained that a total of 8881 vehicles chose to queue and 1499 chose to transfer.

5.2.2. Calculation results under the optimized configuration proportion

The genetic algorithm is used to optimize the solution by adjusting the construction number of charging piles in each parking lot.

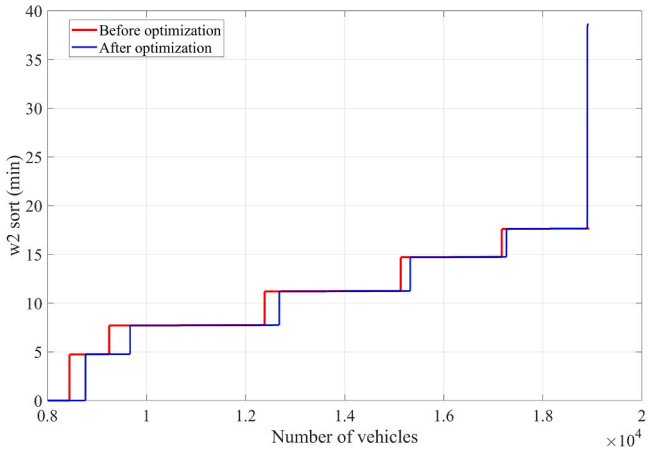


Fig. 6. The time cost $w2$ of each vehicle transferring to the new parking lot before and after optimization.

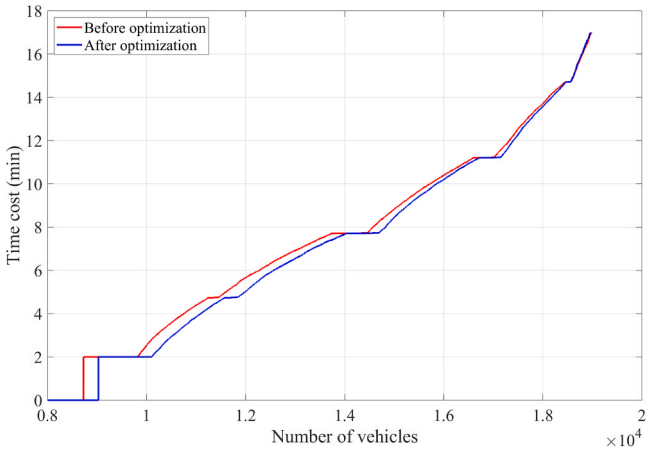


Fig. 7. The total time cost W of each vehicle before and after optimization.

The number of charging piles and parking spaces in each parking lot after optimizing the minimum total time cost is shown in Table 3.

After optimizing the number of charging piles, the time cost of each vehicle is calculated as shown by the blue line in Fig. 7.

We sum the total time cost of each car, and the objective function W is obtained as:

$$W = 7.68 \times 10^4 (\text{min})$$

The W value is reduced by 10.5% compared with the W value under the initial number of charging piles.

We count the estimated queuing time $w1$ and the time cost $w2$ of transferring to the new parking lot calculated by each vehicle in front of the parking lot. The results are shown by the blue line in Fig. 5 and Fig. 6. Within the simulation step size of 1440 steps, a total

Table 3
The initial number of charging piles in each charging station.

Parking lot ID	Parking capacity	Number of fast charging	Number of slow charging	Number of GV spaces
1	200	2	3	195
2	180	1	1	178
3	150	1	2	147
4	160	0	1	159
5	180	2	2	176
6	150	1	2	147
7	150	1	1	148
8	200	2	3	195
9	200	2	3	195
Total	1570	12	18	1540

of 18,745 GVs and EVs were generated, of which 9109 vehicles were parked or charged in the target parking lot according to the initial OD. Compared with 8171 vehicles under the initial charging pile, there is an increase of 11.5%. A total of 9636 vehicles are queued and transferred.

To further analyze vehicles queued and transferred to new parking lots, calculates the difference between w1 and w2. According to statistics, there are 8568 vehicles in the queue, a decrease of 3.5% compared with 8881 vehicles under the initial charging pile's configuration. There are 1068 vehicles transferred and a decrease of 28.8% is found compared to the 1499 vehicles in the initial case.

After optimization, the demand for fast charging piles in the area is only 12. The demand for slow charging piles is only 18. Its total number is 30. There is a reduction of 80% compared with the 153 charging piles obtained from the charging demand forecast.

Assume that the time cost of electric vehicles to queue or transfer to a new charging station is the same as the time cost of fuel vehicles. It can be concluded that redundant charging piles will increase the time cost of GVs and reduce the travel cost of EVs, thus increasing the time cost of all vehicles when the number of EVs is small.

Therefore if the charging piles are built in the parking spaces with a fixed proportion, the time cost of GVs will be greatly increased in the promotion of actual policies, which will be unfavorable to the construction of EV charging piles in public opinion.

5.2.3. Optimization results after increasing the time cost weight of EVs

In fact, from the perspective of the parking lot, maintaining the current proportion will serve more vehicles than making a certain percentage of parking spaces as charging spaces under the condition that the time cost for queuing or transferring to a new charging station for EVs has the same weight as the time cost of GVs. Therefore, governmental support is necessary when the EV is still a small market.

Considering government support, this paper increases the weight of the travel time cost of EVs in the objective function. Since the proportion of EVs is 10% and the number of GVs is 9 times that of EVs, resulting in more queuing time cost for GVs than EVs, there will be a winner-takes-all situation. Considering the positive relationship between the number of vehicles and time cost, to put EVs and GVs in the same position, we increase the weight of travel time cost of EVs to 9 times and simulate again, the simulation results are as follows.

After optimizing the minimum total time cost, the plans obtained for the number of charging piles and parking spaces in each parking lot are shown in Table 4.

After optimization, 70 fast-charging piles and 128 slow charging piles need to be built in the area, and the number of charging piles accounts for 13% of all parking spaces.

We calculate the time cost value W of each vehicle before and after the optimized number of charging piles. The results are shown in the following Fig. 8.

After the travel time cost of EVs was weighted, the time cost of each vehicle under the initial number of charging piles is shown as the red line in Fig. 8. By summing the total time cost of each vehicle, the objective function weighted W is obtained as:

$$W = 9.79 \times 10^4 (\text{min})$$

After optimizing the number of charging piles, the time cost of each vehicle is calculated as shown by the blue line in Fig. 8. By summing the total time cost of each vehicle, the objective function weighted W is obtained as:

$$W = 8.12 \times 10^4 (\text{min})$$

Compared with the W value under the initial number of charging piles, the W value is reduced by 17.1%.

We calculate the queuing time cost $w1$ of each vehicle and the time cost $w2$ of transferring to the new parking lot before and after optimization. The results are shown in Fig. 9 and Fig. 10.

Under the initial number of charging piles, a total of 7235 vehicles are parked or charged in the target parking lot according to the initial OD. After optimizing the number of charging piles, a total of 8106 vehicles are parked or charged in the target parking lot according to the initial OD, with an increase of 12%.

After increasing the travel time cost weight of EVs, to minimize the objective function, the number of charging piles to be built increases from 8% of the overall demand to 13%. After optimizing the number of charging piles, the total vehicle travel cost was reduced from 9.79×10^4 (min) to 8.12×10^4 (min) by 17.1%.

This also shows that by using ABM and genetic algorithm to optimize the configuration of charging piles in the parking lot, travelers can reduce the waiting time for charging and increase the utilization rate of charging piles, thereby reducing the time cost of vehicles and providing travelers with more efficient and convenient charging services., and will also promote the popularization and sustainable development of electric vehicles.

6. Conclusions

This paper mainly establishes an optimization model that optimizes the number of EV charging piles in urban public parking lots to balance the supply and demand of EV charging, enhance the convenience of EV charging, and reduce the waiting time and transfer rate of vehicles that arrive at full parking lots. The ABM used in this study simulates the charging behavior of EVs. In the process of optimizing the configuration of the charging facilities, the genetic algorithm is used to solve the problem. The following conclusions were obtained.

Table 4
Configuration of charging piles in each parking lot after optimization.

Parking lot ID	Parking capacity	Fast charging piles	Slow charging piles	Parking spaces	The proportion of charging piles
1	200	8	8	184	8%
2	180	10	23	147	18%
3	150	6	6	138	8%
4	160	5	24	131	18%
5	180	8	6	166	8%
6	150	9	16	125	17%
7	150	6	8	136	9%
8	200	10	12	178	11%
9	200	8	25	167	17%
Total	1570	70	128	Average	13%

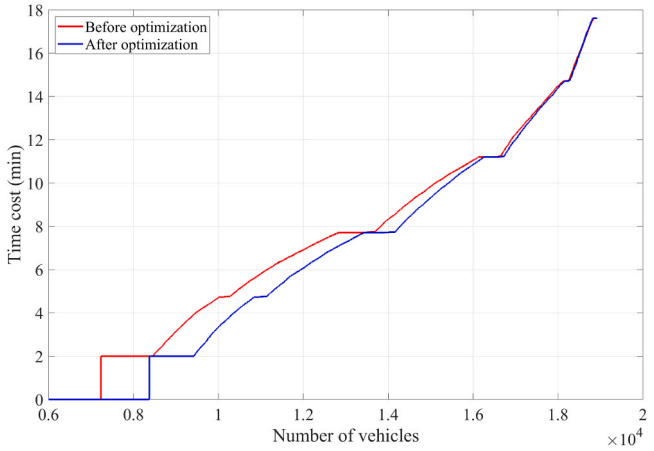


Fig. 8. The total time cost of each vehicle after the travel time cost of EVs was weighted.

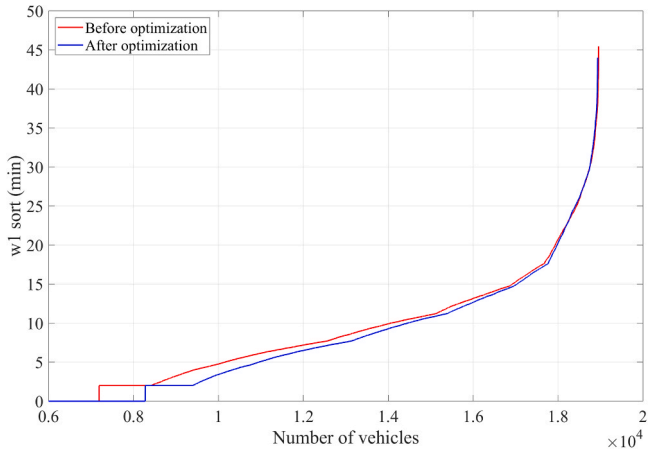


Fig. 9. The value of $w1$ of each vehicle after the travel time cost of EVs was weighted.

(1) This paper addresses the setting of electric vehicle charging piles in public parking lots, and establishes a multi-agent-based simulation model to simulate the driver's behavior and the process of cruising for available charging facilities or available parking spaces. ABM shows good reliability when simulating the coexistence of EV charging and GV parking.
(2) Based on the results of charging demand prediction, this paper proposes to construct an objective function considering the travelers' time cost and obtain the solution by a genetic algorithm to optimize the configuration of charging piles. Feedback and improvement at the micro-level will make the configuration of EV charging facilities coordinated and optimized, which can achieve max system benefits.

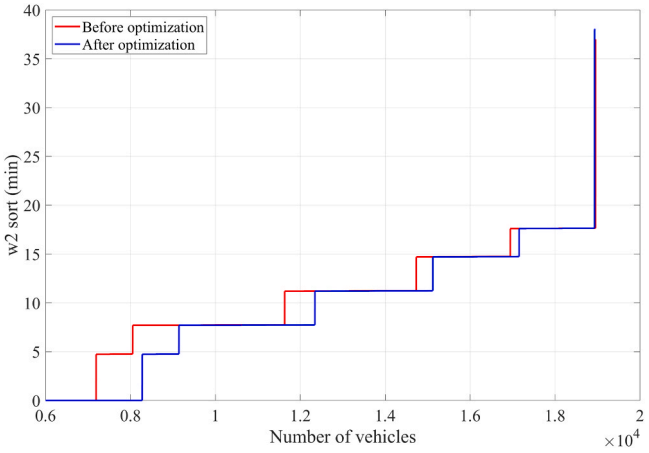


Fig. 10. The value of $w2$ of each vehicle after the travel time cost of EVs was weighted.

(3) This paper takes the abstract road network of a typical CBD in Hangzhou as an example for simulation optimization. The results show that by optimizing the number of charging piles, 17.1% of the total travel time cost of all vehicles can be reduced. The example analysis proves that the agent-based optimization model of EV public parking lot charging station configuration can balance the charging demand of EVs and allocate a more reasonable number of public parking lot charging piles as well as effectively reduce the travel time cost of EVs and GVs. So it can bring a better travel experience to users and improve the overall traffic travel efficiency in the city.

This paper selects the most commonly used mode of rebuilding a certain proportion of parking spaces in the existing parking lot as charging piles. This rate is used as the basis for the subsequent demand forecast and configuration optimization without considering the charging station model only used for charging and private charging piles. At the same time, this paper only considers the travel time cost of travelers in the objective function. The impact of differences in parking lot charges and charging reservations on EV charging choices are not taken into account. These will be studied in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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